
MBAV 443M — Statistical Modeling and AI-Driven Analytics

July 30, 2026 | 2 Hours | 40 Marks

Instructions: Part A (20 marks) is written, answered by hand, scanned, and emailed. **Each Part A question needs only a one-word or one-sentence answer.** Part B (20 marks) is a Python coding task, submitted as a notebook export (PDF/HTML) by email. You have 15 extra minutes after the exam to scan and send. Late submissions are accepted with a mark deduction.

Submission: Email all files to mbav443m@gmail.com by 9:15 PM sharp.

AI Usage Policy: This is a non-proctored exam, so it relies on your honesty. For **Part A (theory)**, AI tools are **not permitted** – answers must reflect your own understanding. For **Part B (coding)**, you **may** use AI tools (e.g., ChatGPT, Copilot, Claude) to assist with writing or debugging code, but the analysis, interpretation, and final recommendation must be your own. Use AI as a coding aid, not a substitute for thinking.

Part A — Written: One Word / One Sentence (20 Marks)

- Q1.** A dataset has many missing values. What should you do before building a predictive model? (2)
- Q2.** Two variables have a correlation of 0.95. What problem can this create in a linear regression model? (2)
- Q3.** A disease detection model misses many actual patients. Which metric should be improved: Precision or Recall? Briefly explain. (2)
- Q4.** A supermarket finds that customers often buy bread and butter together. Which analytical technique would you use to discover such product combinations? (2)
- Q5.** A chatbot can summarize a 20-page report in seconds. What feature of the Transformer model makes this possible? (2)
- Q6.** A retailer has customer data with over 150 variables, making clustering slow and difficult. What should you do before clustering, and why? (2)
- Q7.** A company wants to know whether a new advertisement performs better than the current one. Which statistical analysis would you recommend, and why? (2)
- Q8.** A bank wants to predict whether a loan applicant will default (Yes/No). Which type of analytical model would you recommend, and why? (2)
- Q9.** An online retailer wants to group customers based on similar purchasing behaviour for targeted marketing. Which analytical technique would you recommend, and why? (2)
- Q10.** A prediction model performs extremely well on training data but poorly on new data. What is the most likely reason, and how can it be reduced? (2)

Bonus Question (+2). During this course, we discussed one research paper at least three times across different modules. Name that paper.

Part B — Coding Task (20 Marks)

The attached dataset contains historical records of a Portuguese bank's telemarketing campaign, available from the UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/222/bank+marketing>.

Objective: Develop a predictive model to identify whether a customer will subscribe to a term deposit (Target variable: y).

1. Perform data preprocessing (handle missing values, encode categorical variables, and split the data into training and testing sets). (4 Marks)
2. Build both a Logistic Regression model and a Decision Tree model. (5 Marks)
3. Evaluate the models using Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and ROC Curve. (7 Marks)
4. Recommend which model the bank should deploy and justify your recommendation using the evaluation metrics and business considerations. (4 Marks)

GitHub & Submission Format: Push your code to a public GitHub repository named `mbav443-final-exam`, and mention your GitHub username on the first page of your report. Submit the following by email:

- Your Jupyter notebook (.ipynb)
- Your report (.docx or .pdf), including your GitHub username and repo link
- The dataset (.csv) – **only** if you downloaded it manually; skip this if your notebook imports it directly from the UCI ML Repository

— End of Question Paper —

All the best!